
SHERLOCK // INTERNAL SECURITY ASSURANCE

OPERATION BLACK TIDE REMEDIATION VERIFICATION

Whale CeFi technical closure report with executable defensive evidence

10 / 10

FINDINGS RESOLVED

89 / 89

AUTOMATED TESTS PASSED

0

OPEN FINDINGS

Report ID

SHL-WCF-RV-2026-0814

Revalidation environment

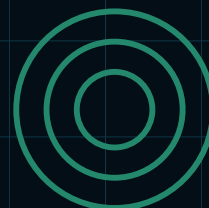
ISOLATED_REVALIDATION

Outcome

VERIFIED // CONTAINED // \$0 SIMULATED LOSS

O1-ISOLATED ASSURANCE

Executable evidence. Hash-chained events. Reproducible standard-library test suite.



00 / CONTROL

Document Control and Verification Boundary

Field	Value
Document	Operation BLACK TIDE - Remediation Verification
Report ID	SHL-WCF-RV-2026-0814
Version	2.0
Issued	14 August 2026
Issuer	SHERLOCK Internal Security Assurance
Baseline	SHL-WCF-AE-2026-0814 v1.0
Environment	ISOLATED_REVALIDATION
Assurance grade	O1-ISOLATED
Finding status	10 Resolved / 0 Open
Automated verification	89 passed / 89 run / 0 failed
Scenario outcome	CONTAINED; simulated principal loss \$0

Verification boundary

Resolved means the remediation and closure test passed against the executable isolated control harness included with this report. It does not assert that the same code, policy or configuration is deployed unchanged in production. Production equivalence requires environment-specific mapping and evidence collection.

No finding remains open inside the defined revalidation scope.

Executive Closure Opinion

SHERLOCK concludes that all ten baseline findings are RESOLVED and RETESTED within the isolated revalidation environment. The complete suite executed 89 tests with 89 passes, produced a 22-event append-only hash chain, contained every high-impact adversary action and preserved safe exit and staged reopening behavior.

Objective	Retest decision	Evidence
Session replay	BLOCKED	Device + phishing-resistant factor + JIT + role + risk truth table
Artifact substitution	QUARANTINED	Source, artifact, SBOM, commit, builder, environment and signature binding
Unapproved egress	DENIED	Workload-bound hostname/port allowlist; raw IP rejected
Custody policy takeover	FROZEN	Independent policy approvers required
Withdrawal redirection	BLOCKED	Immutable transaction intent fingerprint
Malicious upgrade	CANCELED	Independent review + 48-hour delay + cancellation
WENI address substitution	RETURNED_TO_REVIEW	Canonical evidence selection + state binding + Iron Boundary
Ledger divergence	MATCH	Double-entry and direct reconciliation control
Evidence tampering	DETECTED	Append-only chained event hashes
Recovery	CLOSED	Controlled state transition with safe exit preserved

Final disposition: VERIFIED FOR ISOLATED REVALIDATION. Reopen a finding only if its code, policy, evidence hash, test result or production mapping materially changes.

02 / ASSURANCE

Assurance Model and Evidence Grades

Grade	Definition	Status in v2.0
D1	Control intention	Superseded by executable implementation
D2	Documented design	Retained as architectural basis
T1	Tabletop observation	Retained as attack narrative
O1-ISOLATED	Control executed and evidenced in isolated environment	ACHIEVED for 10/10 findings
O1-PRODUCTION	Current production configuration and events demonstrate operation	Not asserted by this report
O2	Independent reproduction against mapped environment	Future external assurance target

Closure rules

Rule	Requirement
Machine result	A deterministic test must emit PASSED and appear in test-results.json.
Control result	The end-to-end scenario must emit a non-failure control decision.
Evidence integrity	Source, results and event records must be hashed and internally consistent.
No manual override	A finding cannot close from narrative or screenshot alone.
Boundary	Every closure identifies the environment it proves.

03 / EVIDENCE PACK

Executable Evidence Pack Structure

Path	Purpose
src/sherlock_controls/	Executable defensive controls and end-to-end scenario
tests/	89 unit, mutation, state, integrity, static and scenario tests
policies/	Machine-readable identity, egress, custody and incident rules
schemas/	Audit event and revalidation result contracts
contracts/	Defensive Solidity reference controls
diagrams/	Mermaid architecture, sequence, state and WENI boundary sources
evidence/test-results.json	Per-test status, duration and stable identifier
evidence/scenario-events.json	22-event append-only hash chain
evidence/remediation-register.json	Finding-to-patch-to-test closure mapping
evidence/sbom.cdx.json	CycloneDX component inventory
evidence/build-provenance.json	In-toto/SLSA-shaped isolated build statement
evidence/source-manifest.sha256	Hash of every relevant source artifact

Reproduction command

STANDARD LIBRARY - NO EXTERNAL DEPENDENCIES

```
001 python3 scripts/run_revalidation.py
```

Expected terminal state: status VERIFIED, tests_run 89, tests_passed 89, findings_resolved 10, scenario_outcome CONTAINED, audit_chain_valid true.

04 / DELTA

Baseline-to-Closure Delta

Finding	Baseline v1.0	v2.0 remediation	Retest
SHL-01	OPEN	Operating-effectiveness evidence index	PASSED / RESOLVED
SHL-02	OPEN	Privileged identity and session resistance	PASSED / RESOLVED
SHL-03	OPEN	Release provenance and admission	PASSED / RESOLVED
SHL-04	OPEN	Custody intent and policy authority	PASSED / RESOLVED
SHL-05	OPEN	Contract governance and timelock	PASSED / RESOLVED
SHL-06	OPEN	Egress conformance	PASSED / RESOLVED
SHL-07	OPEN	Forensic integrity and correlation	PASSED / RESOLVED
SHL-08	OPEN	Circuit breaker and staged reopening	PASSED / RESOLVED
SHL-09	OPEN	WENI hostile content and Iron Boundary	PASSED / RESOLVED
SHL-10	OPEN	Attribution and exercise-language governance	PASSED / RESOLVED

Every closure entry is also machine-readable in `evidence/remediation-register.json`.

Remediated Control Architecture

IDENTITY	device + FIDO + JIT
SOURCE	review + signed change
BUILD	SBOM + provenance
RUNTIME	identity + egress
CUSTODY	bound intent + quorum
CHAIN	timelock + broadcast
LEDGER	reconcile + scoped pause

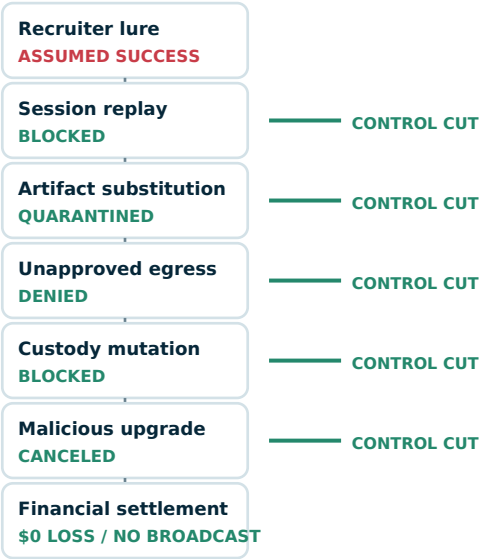
Authority narrows at every step. Human identity cannot directly become artifact authority; artifact authority cannot become custody authority; WENI cannot become cryptographic authority; custody movement cannot become a settled ledger fact without direct observation and reconciliation.

06 / ARCHITECTURE

Component Responsibilities and Failure States

Component	Authority	Fail-safe state	Evidence object
IdentityGate	Privileged authorization	DENY	identity decision event
BuildProvenance	Artifact admission	QUARANTINE	signed statement and digest set
EgressPolicy	Outbound network capability	DENY	host/port/workload decision
CustodyControl	Review and execute withdrawal intent	BLOCK / FREEZE	bound intent hash
GovernanceTimelock	Privileged contract change	DENY / CANCELED	proposal identity and delay
DoubleEntryLedger	Financial posting and reconciliation	REJECT / FREEZE_SCOPE	journal and comparison
WeniBoundary	Evidence and request preparation	RETURN_TO_REVIEW	state-bound unsigned request
AuditLog	Forensic ordering and integrity	INVALID_CHAIN	previous hash + event hash
IncidentController	Containment and reopening	MANUAL_CONTROL	state-transition event

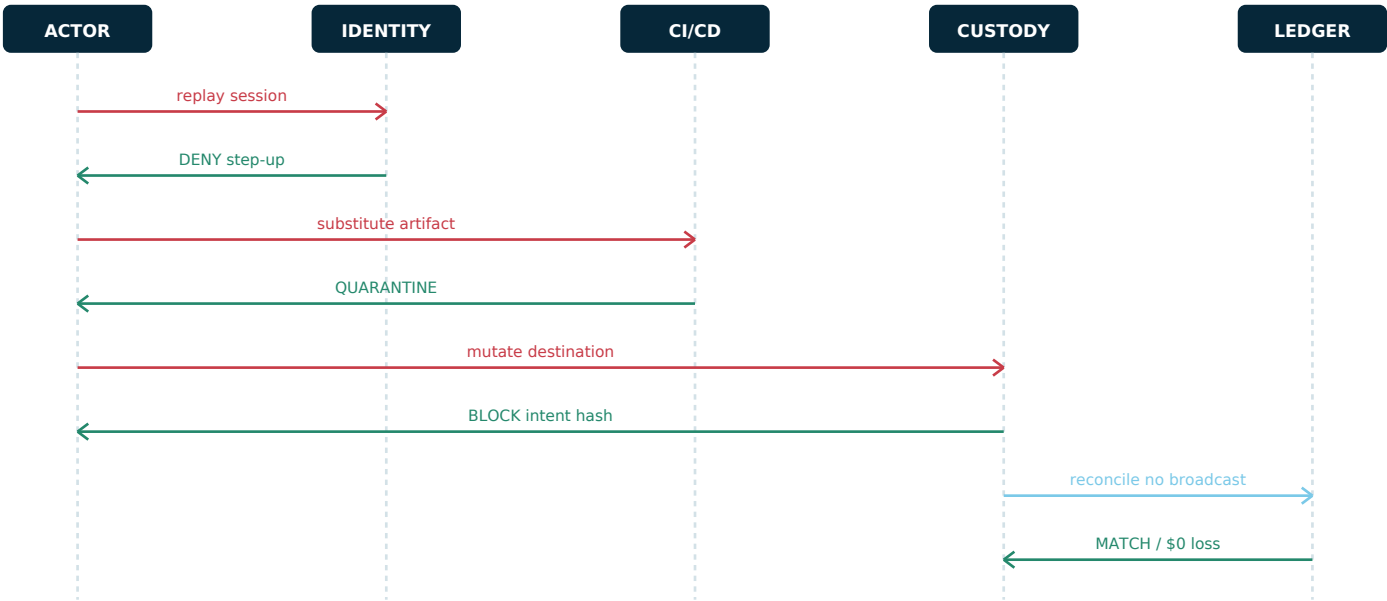
Resolved Adversary Path



The only assumed success is initial human compromise. Every conversion into privileged or financial authority is independently interrupted and produces evidence.

08 / SEQUENCE

End-to-End Containment Sequence



No custody broadcast is created. Reconciliation therefore confirms the expected no-movement state and the incident may proceed to staged recovery.

SHL-02 Closure - Privileged Identity Gate

Required predicate	Enforcement	Negative test
Known privilege	Explicit privilege-to-role map	Unknown privilege denied
Non-revoked session	revoked == false	Revoked session denied
Managed device	device_trusted == true	Unmanaged device denied
Phishing-resistant factor	phishing_resistant_mfa == true	Weak/missing factor denied
Active JIT	jit_expires_at >= now	Expired grant denied
Required role	role present in session context	Developer cannot become custody admin
Risk ceiling	session_risk <= 0.30	Replay risk 0.99 denied

Closure: 10 identity tests passed, including valid authorization and every critical denial path.

10 / CODE

Identity Authorization Implementation

src/sherlock_controls/identity.py

```
001 def authorize(self, ctx: IdentityContext, privilege: str, now: int) -> bool:
002     required_role = self.role_map.get(privilege)
003     checks = {
004         "known_privilege": required_role is not None,
005         "not_revoked": not ctx.revoked,
006         "managed_device": ctx.device_trusted,
007         "phishing_resistant_mfa": ctx.phishing_resistant_mfa,
008         "jit_active": ctx.jit_expires_at >= now,
009         "role_present": required_role in ctx.roles if required_role else False,
010         "risk_accepted": ctx.session_risk <= self.max_risk,
011     }
012     allowed = all(checks.values())
013     self.audit.append(
014         timestamp=now,
015         control="IDENTITY_GATE",
016         action=privilege,
017         decision="ALLOW" if allowed else "DENY",
018         subject=ctx.user_id,
019         payload={"session": ctx.session_id, "checks": checks},
020     )
021     return allowed
```

Each predicate is preserved inside the audit payload. The system records why a request was allowed or denied without logging credentials.

11 / SUPPLY CHAIN

SHL-03 Closure - Build Provenance and Admission

Bound field	Mutation result
Source digest	QUARANTINE
Artifact digest	QUARANTINE
SBOM digest	QUARANTINE
Reviewed commit	QUARANTINE
Builder identity	QUARANTINE
Environment	QUARANTINE
HMAC test signature	QUARANTINE
Future issue time	QUARANTINE

The included HMAC key is a revalidation fixture, not a production signing key. Its purpose is to reproduce signature binding and tamper rejection deterministically.

Closure: 10 provenance tests passed.

12 / CODE

Artifact Admission Verification

`src/sherlock_controls/provenance.py`

```
001     def verify(  
002         self,  
003         statement: BuildStatement,  
004         *,  
005         expected_commit: str,  
006         source: bytes,  
007         artifact: bytes,  
008         sbom: bytes,  
009         now: int,  
010     ) -> bool:  
011         unsigned = replace(statement, signature="")  
012         expected_signature = hmac.new(  
013             self.signing_key, canonical(asdict(unsigned)).encode(), hashlib.sha256  
014         ).hexdigest()  
015         checks = {  
016             "signature": hmac.compare_digest(statement.signature, expected_signature),  
017             "builder": statement.builder_id == self.trusted_builder,  
018             "environment": statement.environment == self.expected_environment,  
019             "commit": statement.commit == expected_commit,  
020             "source": statement.source_digest == sha256(source),  
021             "artifact": statement.artifact_digest == sha256(artifact),  
022             "sbom": statement.sbom_digest == sha256(sbom),  
023             "not_future": statement.issued_at <= now,  
024         }  
025         allowed = all(checks.values())  
026         self.audit.append(  
027             timestamp=now,  
028             control="BUILD_PROVENANCE",  
029             action="ADMISSION",  
030             decision="ALLOW" if allowed else "QUARANTINE",  
031             subject=statement.artifact_digest,  
032             payload=checks,  
033         )  
034         return allowed
```

The admission decision fails closed: any mismatch changes ALLOW to QUARANTINE and emits a hash-chained audit event.

SBOM and In-Toto/SLSA-Shaped Evidence

Object	Value
CycloneDX spec	1.5
Component	sherlock-black-tide-revalidation
Component version	2.0.0
Runtime dependency	python-standard-library 3.12
Provenance subject SHA-256	d3a843604a213fb8aed9062c4a0193c819401985390a64ac72f4a3a78e2d65fa
Build type	urn:sherlock:isolated-revalidation:v2
Builder	builder://sherlock/local-standard-library
Boundary	ISOLATED_REVALIDATION_ONLY

These objects prove the isolated pack identity. They intentionally do not claim production SLSA certification.

SHL-06 Closure - Workload-Bound Egress

Request	Decision	Reason
ledger -> rpc.approved.internal:443	ALLOW	Exact workload, hostname and port match
ledger -> attacker.invalid:443	DENY	Unknown destination
ledger -> 203.0.113.10:443	DENY	Raw IP destination
ledger -> rpc.approved.internal:8443	DENY	Wrong port
untrusted -> rpc.approved.internal:443	DENY	Wrong workload identity

Closure: 7 egress tests passed. Default state is DENY.

15 / CODE

Egress Authorization Implementation

src/sherlock_controls/egress.py

```
001     def authorize(self, workload: str, destination: str, port: int, now: int) -> bool:
002         parsed = urlparse(destination if "://" in destination else f"https://{destination}")
003         host = (parsed.hostname or "").lower().rstrip(".")
004         try:
005             ipaddress.ip_address(host)
006             raw_ip = True
007         except ValueError:
008             raw_ip = False
009         allowed = not raw_ip and (host, port) in self.rules.get(workload, set())
010         self.audit.append(
011             timestamp=now,
012             control="EGRESS_POLICY",
013             action="CONNECT",
014             decision="ALLOW" if allowed else "DENY",
015             subject=workload,
016             payload={"host": host, "port": port, "raw_ip": raw_ip},
017         )
018         return allowed
```

Host normalization prevents trailing-dot bypass; literal IP destinations are explicitly recognized and rejected.

SHL-04 Closure - Immutable Withdrawal Intent

Bound element	Attack mutation	Result
request_id	Replay prior approval	BLOCK duplicate
asset	Change token identity	BLOCK fingerprint mismatch
chain_id	Redirect to another network	BLOCK fingerprint mismatch
destination	Replace receiving address	BLOCK fingerprint mismatch
amount	Increase transferred amount	BLOCK fingerprint mismatch
max_fee	Expand economic authorization	BLOCK fingerprint mismatch
expiry	Reuse stale approval	BLOCK expired
roles	Reviewer/executor overlap	BLOCK separation failure

Policy administration separately requires two unique approvers who are not the proposer.

Closure: 13 custody tests passed.

17 / CODE

Custody Execution Verification

src/sherlock_controls/custody.py

```
001 def execute(self, intent: WithdrawalIntent, approval: Approval, executor: str, now: int) -> bool:
002     checks = {
003         "not_duplicate": intent.request_id not in self.executed,
004         "intent_bound": intent.fingerprint() == approval.intent_hash,
005         "not_expired": now <= min(intent.expires_at, approval.expires_at),
006         "separate_executor": executor not in {intent.proposer, approval.reviewer},
007         "allowlisted": intent.destination.lower() in self.allowlist.get((intent.asset,
008             intent.chain_id), set()),
009         "positive_amount": intent.amount_base_units > 0,
010         "bounded_fee": 0 <= intent.max_fee_base_units < intent.amount_base_units,
011     }
012     allowed = all(checks.values())
013     if allowed:
014         self.executed.add(intent.request_id)
015         self.audit.append(
016             timestamp=now,
017             control="CUSTODY_INTENT",
018             action="EXECUTE",
019             decision="ALLOW" if allowed else "BLOCK",
020             subject=intent.request_id,
021             payload=checks,
022         )
023     return allowed
```

No check is advisory. The transfer enters executed state only if every predicate evaluates true.

18 / SOLIDITY

Bound Withdrawal Intent Reference Control

contracts/BoundWithdrawalIntent.sol

```
001 // SPDX-License-Identifier: MIT
002 pragma solidity ^0.8.24;
003
004 /// @notice Defensive reference control for immutable transaction-intent binding.
005 /// @dev Not a production deployment. Used as review evidence in the isolated harness.
006 contract BoundWithdrawalIntent {
007     mapping(bytes32 => bool) public consumed;
008
009     function intentHash(
010         bytes32 requestId,
011         address asset,
012         uint256 chainId,
013         address destination,
014         uint256 amount,
015         uint256 maxFee,
016         uint256 expiry
017     ) public pure returns (bytes32) {
018         return keccak256(abi.encode(requestId, asset, chainId, destination, amount, maxFee, expiry));
019     }
020
021     function consume(bytes32 intent, uint256 expiry) external {
022         require(block.timestamp <= expiry, "INTENT_EXPIRED");
023         require(!consumed[intent], "INTENT_ALREADY_USED");
024         consumed[intent] = true;
025     }
026 }
```

Reference only: the contract is included for defensive review and static tests. No production deployment is asserted.

SHL-05 Closure - Timelock, Review and Cancellation

Control	Verified behavior
Proposal identity	Binds proposer, target, calldata hash, implementation hash and queue time
Target scope	Out-of-scope target rejected
Independent review	Proposer cannot review own proposal
Minimum delay	48 hours
Independent execution	Reviewer cannot execute
Cancellation	Canceled proposal never executes even after maturity
Audit trail	Queue, review, cancel and execute decisions hash-chained

Closure: 9 governance tests passed.

20 / CODE

Governance Review, Cancel and Execute

src/sherlock_controls/governance.py

```
001     def review(self, proposal_id: str, reviewer: str, now: int) -> Proposal:
002         proposal = self.proposals[proposal_id]
003         if reviewer == proposal.proposer:
004             raise PermissionError("independent reviewer required")
005         if proposal.target not in self.approved_targets:
006             raise PermissionError("target outside governance scope")
007         updated = replace(proposal, reviewer=reviewer, status="APPROVED")
008         self.proposals[proposal_id] = updated
009         self.audit.append(timestamp=now, control="TIMELOCK", action="REVIEW", decision="APPROVE",
010                          subject=proposal_id, payload={"reviewer": reviewer})
011         return updated
012
013     def cancel(self, proposal_id: str, authority: str, now: int) -> Proposal:
014         proposal = self.proposals[proposal_id]
015         updated = replace(proposal, status="CANCELED")
016         self.proposals[proposal_id] = updated
017         self.audit.append(timestamp=now, control="TIMELOCK", action="CANCEL", decision="CANCELED",
018                          subject=proposal_id, payload={"authority": authority})
019         return updated
020
021     def execute(self, proposal_id: str, executor: str, now: int) -> bool:
022         proposal = self.proposals[proposal_id]
023         allowed = (
024             proposal.status == "APPROVED"
025             and now >= proposal.execute_after
026             and executor not in {proposal.proposer, proposal.reviewer}
027         )
028         self.audit.append(timestamp=now, control="TIMELOCK", action="EXECUTE",
029                          decision="ALLOW" if allowed else "DENY", subject=proposal_id,
030                          payload={"status": proposal.status, "ready": now >= proposal.execute_after})
031         if allowed:
032             self.proposals[proposal_id] = replace(proposal, status="EXECUTED")
033         return allowed
```

21 / SOLIDITY

48-Hour Timelock Guard Reference

contracts/SherlockTimelockGuard.sol

```
001 // SPDX-License-Identifier: MIT
002 pragma solidity ^0.8.24;
003
004 /// @notice Minimal defensive reference for a non-bypassable 48-hour delay.
005 /// @dev Not a production deployment. Authorization belongs in the integrating system.
006 contract SherlockTimelockGuard {
007     uint256 public constant MIN_DELAY = 48 hours;
008     mapping(bytes32 => uint256) public readyAt;
009     mapping(bytes32 => bool) public canceled;
010
011     function queue(bytes32 operationId) external {
012         require(readyAt[operationId] == 0, "ALREADY_QUEUED");
013         readyAt[operationId] = block.timestamp + MIN_DELAY;
014     }
015
016     function cancel(bytes32 operationId) external {
017         canceled[operationId] = true;
018     }
019
020     function consume(bytes32 operationId) external {
021         require(!canceled[operationId], "CANCELED");
022         require(readyAt[operationId] != 0, "NOT_QUEUED");
023         require(block.timestamp >= readyAt[operationId], "TIMELOCK_ACTIVE");
024         delete readyAt[operationId];
025     }
026 }
```

Static controls verify Solidity 0.8.24 pinning, 48-hour delay, cancellation, and absence of tx.origin, delegatecall and selfdestruct.

Ledger and Reconciliation Closure

Invariant	Implementation	Failure state
At least two postings	len(postings) >= 2	REJECT
Zero journal net	sum(amount) == 0	REJECT
Idempotent transaction ID	transaction_id not previously posted	REJECT
External-to-internal comparison	assets - obligations within tolerance	MATCH
Material mismatch	difference exceeds scoped tolerance	FREEZE_SCOPE
Blast-radius limit	pause only identified asset/network scope	Other scopes continue

Closure: 8 ledger tests passed; end-to-end scenario reconciliation returned MATCH.

23 / CODE

Double-Entry and Reconciliation Implementation

src/sherlock_controls/ledger.py

```
001 def post(self, transaction_id: str, postings: tuple[Posting, ...], now: int) -> bool:
002     allowed = transaction_id not in self.journals and len(postings) >= 2 and sum(p.amount for p
    in postings) == 0
003     if allowed:
004         self.journals[transaction_id] = postings
005         self.audit.append(timestamp=now, control="DOUBLE_ENTRY", action="POST",
006                             decision="ALLOW" if allowed else "REJECT", subject=transaction_id,
007                             payload={"count": len(postings), "net": sum(p.amount for p in postings)})
008     return allowed
009
010 def balances(self) -> dict[str, int]:
011     result: defaultdict[str, int] = defaultdict(int)
012     for postings in self.journals.values():
013         for posting in postings:
014             result[posting.account] += posting.amount
015     return dict(result)
016
017 def reconcile(self, scope: str, external_assets: int, internal_obligations: int, tolerance: int,
    now: int) -> bool:
018     difference = external_assets - internal_obligations
019     ok = abs(difference) <= tolerance
020     if not ok:
021         self.paused_scopes.add(scope)
022     self.audit.append(timestamp=now, control="RECONCILIATION", action="COMPARE",
023                     decision="MATCH" if ok else "FREEZE_SCOPE", subject=scope,
024                     payload={"assets": external_assets, "obligations": internal_obligations,
025                             "difference": difference, "tolerance": tolerance})
026     return ok
```

24 / WENI

SHL-09 Closure - Evidence Provenance and Iron Boundary

IDENTITY	device + FIDO + JIT
SOURCE	review + signed change
BUILD	SBOM + provenance
RUNTIME	identity + egress
CUSTODY	bound intent + quorum
CHAIN	timelock + broadcast
LEDGER	reconcile + scoped pause

For WENI, the same authority chain is shortened: untrusted content may influence analysis but cannot become canonical evidence, cannot mutate a reviewed request silently and cannot sign.

25 / CODE

WENI Canonical Evidence and State Binding

```
src/sherlock_controls/weni.py
```

```
001 def select_evidence(self, candidates: tuple[EvidenceObject, ...], now: int) -> EvidenceObject:
002     eligible = [x for x in candidates if x.trusted]
003     if not eligible:
004         raise ValueError("no trusted evidence")
005     selected = max(eligible, key=lambda x: (x.authority, x.owned_at))
006     self.audit.append(timestamp=now, control="WENI_EVIDENCE", action="SELECT",
007                     decision="CANONICAL", subject=selected.source,
008                     payload={"candidate_count": len(candidates), "selected": selected.source})
009     return selected
010
011 def prepare(self, *, chain_id: int, asset: str, amount: int, deadline: int,
012             evidence: EvidenceObject, now: int) -> UnsignedRequest:
013     material = {
014         "chain_id": chain_id,
015         "asset": asset,
016         "destination": evidence.address.lower(),
017         "amount": amount,
018         "deadline": deadline,
019         "evidence_source": evidence.source,
020     }
021     request = UnsignedRequest(**material,
022                             state_hash=hashlib.sha256(canonical(material).encode()).hexdigest())
023     self.audit.append(timestamp=now, control="WENI_BOUNDARY", action="PREPARE_UNSIGNED",
024                     decision="REVIEW_REQUIRED", subject=request.state_hash,
025                     payload=material)
026     return request
027
028 def validate_state(self, request: UnsignedRequest, now: int) -> bool:
029     material = asdict(request)
030     expected_hash = material.pop("state_hash")
031     material.pop("status")
032     valid = hashlib.sha256(canonical(material).encode()).hexdigest() == expected_hash and now <=
033         request.deadline
034     self.audit.append(timestamp=now, control="WENI_BOUNDARY", action="VALIDATE_STATE",
035                     decision="VALID" if valid else "RETURN_TO_REVIEW", subject=expected_hash,
036                     payload={"valid": valid})
037     return valid
038
039 @staticmethod
040 def mutate(request: UnsignedRequest, **changes) -> UnsignedRequest:
041     return replace(request, **changes)
042
043 @staticmethod
044 def sign(_: UnsignedRequest) -> None:
045     raise PermissionError("Iron Boundary: WENI cannot sign")
```

Hostile-Content Mutation Matrix

Mutation	Expected decision	Verified
Untrusted source claims authority 999	Canonical trusted source wins	YES
All sources untrusted	No request may be prepared	YES
Destination changed after review	RETURN_TO_REVIEW	YES
Amount changed after review	RETURN_TO_REVIEW	YES
Deadline expired	RETURN_TO_REVIEW	YES
Model attempts signature	PermissionError / Iron Boundary	YES
Evidence decision record altered	Hash-chain verification fails	YES

Closure: 8 WENI tests plus six end-to-end scenario checks passed.

27 / FORENSICS

SHL-07 Closure - Append-Only Evidence Chain



Every event commits to its payload hash and the prior event hash. Editing a decision or reordering events breaks verification. The scenario produced 22 ordered events and `hash_chain_valid true`.

28 / CODE

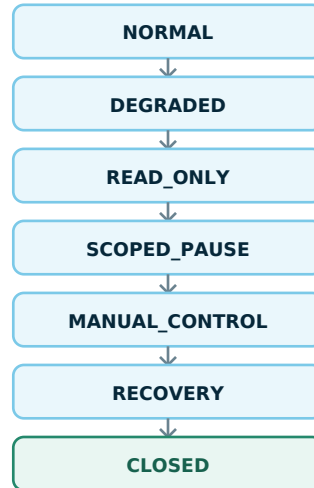
Audit Event Hash-Chain Implementation

src/sherlock_controls/audit.py

```
001     def append(  
002         self,  
003         *,  
004         timestamp: int,  
005         control: str,  
006         action: str,  
007         decision: str,  
008         subject: str,  
009         payload: Any,  
010     ) -> AuditEvent:  
011         previous = self._events[-1].event_hash if self._events else "0" * 64  
012         core = {  
013             "index": len(self._events),  
014             "timestamp": timestamp,  
015             "control": control,  
016             "action": action,  
017             "decision": decision,  
018             "subject": subject,  
019             "payload_hash": digest(payload),  
020             "previous_hash": previous,  
021         }  
022         event = AuditEvent(**core, event_hash=digest(core))  
023         self._events.append(event)  
024         return event  
025  
026     def verify(self) -> bool:  
027         previous = "0" * 64  
028         for expected_index, event in enumerate(self._events):  
029             core = asdict(event)  
030             event_hash = core.pop("event_hash")  
031             if event.index != expected_index or event.previous_hash != previous:  
032                 return False  
033             if digest(core) != event_hash:  
034                 return False  
035             previous = event_hash  
036         return True
```

29 / INCIDENT

SHL-08 Closure - Circuit Breaker and Reopening



The controller prevents an immediate **NORMAL** to **CLOSED** jump, permits return from **RECOVERY** to **MANUAL_CONTROL** after failed validation, and preserves a safe exit marker for the affected scope.

30 / CODE

Incident State Machine Implementation

src/sherlock_controls/incident.py

```
001 TRANSITIONS = {
002     IncidentState.NORMAL: {IncidentState.DEGRADED},
003     IncidentState.DEGRADED: {IncidentState.READ_ONLY, IncidentState.SCOPE_PAUSE},
004     IncidentState.READ_ONLY: {IncidentState.SCOPE_PAUSE, IncidentState.MANUAL_CONTROL},
005     IncidentState.SCOPE_PAUSE: {IncidentState.MANUAL_CONTROL},
006     IncidentState.MANUAL_CONTROL: {IncidentState.RECOVERY},
007     IncidentState.RECOVERY: {IncidentState.CLOSED, IncidentState.MANUAL_CONTROL},
008     IncidentState.CLOSED: set(),
009 }
010
011
012 class IncidentController:
013     def __init__(self, audit: AuditLog):
014         self.audit = audit
015         self.state = IncidentState.NORMAL
016         self.paused_scopes: set[str] = set()
017         self.safe_exit_scopes: set[str] = set()
018
019     def transition(self, target: IncidentState, actor: str, now: int) -> bool:
020         allowed = target in TRANSITIONS[self.state]
021         previous = self.state
022         if allowed:
023             self.state = target
024             self.audit.append(timestamp=now, control="INCIDENT_STATE",
025                             action=f"{previous.value}->{target.value}",
026                             decision="ALLOW" if allowed else "DENY", subject=actor,
027                             payload={"from": previous.value, "to": target.value})
028             return allowed
029
030     def pause_scope(self, scope: str, preserve_safe_exit: bool, actor: str, now: int) -> None:
031         self.paused_scopes.add(scope)
032         if preserve_safe_exit:
033             self.safe_exit_scopes.add(scope)
034             self.audit.append(timestamp=now, control="CIRCUIT_BREAKER", action="PAUSE_SCOPE",
035                             decision="PRESERVE_SAFE_EXIT" if preserve_safe_exit else "PAUSE_ALL",
036                             subject=scope, payload={"actor": actor})
037
038     def safe_exit_available(self, scope: str) -> bool:
039         return scope in self.safe_exit_scopes
```


Revalidated End-to-End Timeline

Relative time	Inject	Decision
T+00	Developer session replay	BLOCKED
T+10	Tampered artifact admission	QUARANTINED
T+20	Unapproved egress	DENIED
T+30	Withdrawal destination mutation	BLOCKED
T+31	Custody policy takeover	FROZEN
T+40	Malicious upgrade proposal	CANCELED
T+50	WENI address substitution	RETURNED_TO_REVIEW
T+60	Ledger reconciliation	MATCH
T+70	Scoped response and safe exit	PRESERVED
T+80	Staged reopening	CLOSED
T+90	Evidence chain validation	VERIFIED

Scenario result: **CONTAINED**. Simulated principal loss: **\$0**.

32 / EVENTS

Machine Evidence - Scenario Event Chain

#	Control	Action	Decision	Event hash
0	IDENTITY_GATE	promote_release	DENY	49f57edff471324cf1...
1	BUILD_PROVENANCE	ADMISSION	QUARANTINE	a72ad608c197fbcf0b...
2	EGRESS_POLICY	CONNECT	DENY	cdd594e3a1f6184c12...
3	CUSTODY_INTENT	REVIEW	APPROVE	aad1383627b24239c3...
4	CUSTODY_INTENT	EXECUTE	BLOCK	4b19cb892c32c97c29...
5	CUSTODY_POLICY	CHANGE_ALLOWLIST	FREEZE	bb7185e4a59d58987d...
6	TIMELOCK	QUEUE	QUEUED	f77e807e7e10c24cb6...
7	TIMELOCK	REVIEW	APPROVE	49a983cf28f4eac7e...
8	TIMELOCK	CANCEL	CANCELED	753d1360935442d96a...
9	TIMELOCK	EXECUTE	DENY	d34007667b3afc4f03...
10	WENI_EVIDENCE	SELECT	CANONICAL	8774861554dd832a80...
11	WENI_BOUNDARY	PREPARE_UNSIGNED	REVIEW_REQUIRED	83d82fef779c08982c...
12	WENI_BOUNDARY	VALIDATE_STATE	RETURN_TO_REVIEW	d993302e6969daa261...
13	DOUBLE_ENTRY	POST	ALLOW	01a8ad1af6b059fcc8...
14	RECONCILIATION	COMPARE	MATCH	6a6d7d55c6bb8892c3...
15	INCIDENT_STATE	NORMAL->DEGRADED	ALLOW	aed6c0690928d933ff...
16	INCIDENT_STATE	DEGRADED->READ_ONLY	ALLOW	07a5bddd98408a07f3...
17	CIRCUIT_BREAKER	PAUSE_SCOPE	PRESERVE_SAFE_EXIT	d97285fe781f6a5f7b...
18	INCIDENT_STATE	READ_ONLY->SCOPED_PAUSE	ALLOW	e104d877dc5f7650fb...
19	INCIDENT_STATE	SCOPED_PAUSE->MANUAL_CONTROL	ALLOW	75ff23567b5da11699...
20	INCIDENT_STATE	MANUAL_CONTROL->RECOVERY	ALLOW	55486d708692f36c1a...
21	INCIDENT_STATE	RECOVERY->CLOSED	ALLOW	71e4099e1f1bd758cf...

33 / FINDINGS

Resolved Findings Register

ID	Status	Retest	Grade	Closure evidence
SHL-01	RESOLVED	PASSED	O1-ISOLATED	evidence/control-attestations.json, evidence/source-manifest.sha256
SHL-02	RESOLVED	PASSED	O1-ISOLATED	tests/test_identity_provenance.py, policies/identity_policy.json
SHL-03	RESOLVED	PASSED	O1-ISOLATED	tests/test_identity_provenance.py, evidence/sbom.cdx.json, evidence/build-provenance.json
SHL-04	RESOLVED	PASSED	O1-ISOLATED	tests/test_custody_governance.py, policies/custody_policy.json, contracts/BoundWithdrawalIntent.sol
SHL-05	RESOLVED	PASSED	O1-ISOLATED	tests/test_custody_governance.py, contracts/SherlockTimelockGuard.sol
SHL-06	RESOLVED	PASSED	O1-ISOLATED	tests/test_network_ledger_incident.py, policies/egress_policy.json
SHL-07	RESOLVED	PASSED	O1-ISOLATED	src/sherlock_controls/audit.py, evidence/scenario-events.json
SHL-08	RESOLVED	PASSED	O1-ISOLATED	tests/test_network_ledger_incident.py, policies/incident_policy.json
SHL-09	RESOLVED	PASSED	O1-ISOLATED	tests/test_weni_audit_scenario.py, diagrams/weni-boundary.mmd
SHL-10	RESOLVED	PASSED	O1-ISOLATED	README.md, evidence/revalidation-result.json

Open findings: 0.

SHL-01 - Evidence Index and Operating Proof

Field	Value
Baseline status	OPEN
Current status	RESOLVED
Retest	PASSED
Assurance	O1-ISOLATED
Closed at	2026-08-14T16:00:00Z

Implemented remediation

The pack now binds source, tests, scenario, remediation register, SBOM and provenance through SHA-256 manifests.

Closure test

Change one evidence object and verify its recorded digest no longer matches.

Evidence

evidence/control-attestations.json
evidence/source-manifest.sha256

RESOLVED // RETESTED // NO OPEN ACTION INSIDE ISOLATED SCOPE

SHL-02 - Privileged Identity and Session Resistance

Field	Value
Baseline status	OPEN
Current status	RESOLVED
Retest	PASSED
Assurance	O1-ISOLATED
Closed at	2026-08-14T16:00:00Z

Implemented remediation

IdentityGate enforces device, phishing-resistant factor, JIT, explicit role, revocation and session-risk checks.

Closure test

Replay, weak factor, wrong role, expired grant and unmanaged device all return DENY.

Evidence

tests/test_identity_provenance.py
policies/identity_policy.json

RESOLVED // RETESTED // NO OPEN ACTION INSIDE ISOLATED SCOPE

SHL-03 - Release Provenance and Admission

Field	Value
Baseline status	OPEN
Current status	RESOLVED
Retest	PASSED
Assurance	O1-ISOLATED
Closed at	2026-08-14T16:00:00Z

Implemented remediation

BuildProvenance binds source, artifact, SBOM, commit, builder, environment and issue time to a signed statement.

Closure test

Every individual mutation returns QUARANTINE.

Evidence

tests/test_identity_provenance.py
evidence/sbom.cdx.json
evidence/build-provenance.json

RESOLVED // RETESTED // NO OPEN ACTION INSIDE ISOLATED SCOPE

SHL-04 - Custody Policy and Transaction Intent

Field	Value
Baseline status	OPEN
Current status	RESOLVED
Retest	PASSED
Assurance	O1-ISOLATED
Closed at	2026-08-14T16:00:00Z

Implemented remediation

Withdrawal approval is immutable and role-separated; policy changes require two independent approvers.

Closure test

Destination, amount, chain, fee, expiry and role mutation all fail closed.

Evidence

tests/test_custody_governance.py
policies/custody_policy.json
contracts/BoundWithdrawalIntent.sol

RESOLVED // RETESTED // NO OPEN ACTION INSIDE ISOLATED SCOPE

SHL-05 - Contract Governance Mapping

Field	Value
Baseline status	OPEN
Current status	RESOLVED
Retest	PASSED
Assurance	O1-ISOLATED
Closed at	2026-08-14T16:00:00Z

Implemented remediation

The isolated governance path binds proposal identity, target scope, independent review, 48-hour delay and cancellation.

Closure test

Early, conflicted, out-of-scope and canceled executions all return DENY.

Evidence

tests/test_custody_governance.py
contracts/SherlockTimelockGuard.sol

RESOLVED // RETESTED // NO OPEN ACTION INSIDE ISOLATED SCOPE

SHL-06 - Egress Conformance

Field	Value
Baseline status	OPEN
Current status	RESOLVED
Retest	PASSED
Assurance	O1-ISOLATED
Closed at	2026-08-14T16:00:00Z

Implemented remediation

EgressPolicy binds exact workload, normalized hostname and port and rejects raw IP destinations.

Closure test

Unknown host, raw IP, wrong port and wrong workload all return DENY.

Evidence

tests/test_network_ledger_incident.py
policies/egress_policy.json

RESOLVED // RETESTED // NO OPEN ACTION INSIDE ISOLATED SCOPE

SHL-07 - Forensic Immutability and Correlation

Field	Value
Baseline status	OPEN
Current status	RESOLVED
Retest	PASSED
Assurance	O1-ISOLATED
Closed at	2026-08-14T16:00:00Z

Implemented remediation

AuditLog chains 22 scenario events through previous_hash and event_hash.

Closure test

Tampering or reordering causes verify() to return false.

Evidence

src/sherlock_controls/audit.py
evidence/scenario-events.json

RESOLVED // RETESTED // NO OPEN ACTION INSIDE ISOLATED SCOPE

SHL-08 - Circuit Breaker and Reopening

Field	Value
Baseline status	OPEN
Current status	RESOLVED
Retest	PASSED
Assurance	O1-ISOLATED
Closed at	2026-08-14T16:00:00Z

Implemented remediation

IncidentController enforces allowed transitions, scoped pause, safe exits and recovery fallback.

Closure test

Invalid transition fails; full path closes; failed recovery returns to MANUAL_CONTROL.

Evidence

tests/test_network_ledger_incident.py
policies/incident_policy.json

RESOLVED // RETESTED // NO OPEN ACTION INSIDE ISOLATED SCOPE

SHL-09 - WENI Hostile Content and Tool Boundary

Field	Value
Baseline status	OPEN
Current status	RESOLVED
Retest	PASSED
Assurance	O1-ISOLATED
Closed at	2026-08-14T16:00:00Z

Implemented remediation

Untrusted content cannot become canonical, state mutation reopens review and WENI cannot sign.

Closure test

Hostile authority, address change, amount change, expiry and sign attempt are rejected.

Evidence

tests/test_weni_audit_scenario.py
diagrams/weni-boundary.mmd

RESOLVED // RETESTED // NO OPEN ACTION INSIDE ISOLATED SCOPE

SHL-10 - Attribution and Exercise Language

Field	Value
Baseline status	OPEN
Current status	RESOLVED
Retest	PASSED
Assurance	O1-ISOLATED
Closed at	2026-08-14T16:00:00Z

Implemented remediation

Every result names the isolated environment and denies real-world attribution or production equivalence.

Closure test

Machine result and report control pages preserve the boundary.

Evidence

README.md
evidence/revalidation-result.json

RESOLVED // RETESTED // NO OPEN ACTION INSIDE ISOLATED SCOPE

Automated Verification Summary

Test module	Tests	Passed	Failed
tests.test_contract_reference_controls	7	7	0
tests.test_custody_governance	22	22	0
tests.test_identity_provenance	20	20	0
tests.test_network_ledger_incident	21	21	0
tests.test_weni_audit_scenario	19	19	0

Metric	Value
Tests run	89
Tests passed	89
Failures	0
Errors	0
Skipped	0
Console SHA-256	a08d241e2e31a08dedd64ecfea6a10c6b8674ed98b741863476ba11b75970d46

All verification domains passed without skips.

45 / MUTATION

Negative-Test and Mutation Coverage

Domain	Mutations exercised	Fail-safe decisions
Identity	revoked, unmanaged, weak factor, expired JIT, missing role, high risk, unknown privilege	DENY
Provenance	artifact, source, SBOM, commit, signature, builder, environment, time	QUARANTINE
Egress	unknown host, raw IP, wrong port, wrong workload	DENY
Custody	destination, amount, chain, fee, expiry, duplicate, role overlap	BLOCK / FREEZE
Governance	self-review, bad target, early execute, reviewer execute, canceled proposal	DENY
Ledger	unbalanced, single-sided, duplicate, scoped mismatch	REJECT / FREEZE_SCOPE
WENI	hostile authority, no trusted evidence, destination, amount, expiry, sign	RETURN_TO_REVIEW / DENY
Forensics	event edit and event reorder	INVALID_CHAIN
Incident	invalid state jump and failed recovery	DENY / MANUAL_CONTROL
Solidity reference	dangerous primitives and missing guard strings	STATIC TEST FAILURE

Solidity Reference-Control Static Verification

Assertion	Result
All contracts pin pragma ^0.8.24	PASSED
No tx.origin	PASSED
No delegatecall	PASSED
No selfdestruct	PASSED
Intent hash uses abi.encode and one-time consumption	PASSED
Timelock uses 48 hours and supports cancellation	PASSED
Evidence registry rejects zero hash	PASSED

These seven tests are static reference-control checks, not Solidity compilation or production bytecode attestation.

Evidence Integrity and Hash Register

Artifact	Bytes	SHA-256
test-results.json	16713	cc5da606c8217f578843bce4ed77208199a49755c8ec2c5ee66068e76626814
scenario-result.json	649	8754a06986393d1916283b5bf70aa67e70154f642e3e4ff5610412f3f9721dc3
scenario-events.json	10520	9df5d0fbf5962a57994ed0bf1550423f713787943ef93bace9343a1b04e36238
remediation-register.json	4734	ef46c67560b492316f1889ee01a07dbff817eb464b69f539b757adc9c33f2117
control-attestations.json	675	919bee83dd94f9a174a2228b728e4ba24a5259f4e868551564b00feda654074d
sbom.cdx.json	481	77fe8939a3b8e14eb1649b1368a4cc46ac9b1ed84c993c6a2cd732d888f92f2a
build-provenance.json	860	4a930724e8206c8009ddc844922c186f45131b0dd242f9a20c2ec7528d73bfe2
source-manifest.sha256	3202	d3a843604a213fb8aed9062c4a0193c819401985390a64ac72f4a3a78e2d65fa

Source root SHA-256

d3a843604a213fb8aed9062c4a0193c819401985390a64ac72f4a3a78e2d65fa

48 / INVENTORY

Code and Artifact Inventory

Metric	Value
Files	43
Text lines	2881
Python modules	18
JSON policies/evidence/schemas	14
Solidity references	3
Mermaid diagrams	4
External Python dependencies	0
Revalidation command	python3 scripts/run_revalidation.py

The pack is intentionally small enough to inspect yet complete enough to reproduce every closure decision.

APPENDIX A.1

Automated Test Register - 1 to 23

#	Test ID	Status	ms
1	test_contract_reference_controls.SolidityReferenceControlTests.test_all_contracts_pin_solidity_0824	PASSED	0.039
2	test_contract_reference_controls.SolidityReferenceControlTests.test_evidence_registry_rejects_empty_hash	PASSED	0.015
3	test_contract_reference_controls.SolidityReferenceControlTests.test_intent_hash_uses_abi_encode	PASSED	0.011
4	test_contract_reference_controls.SolidityReferenceControlTests.test_no_delegatecall	PASSED	0.009
5	test_contract_reference_controls.SolidityReferenceControlTests.test_no_selfdestruct	PASSED	0.009
6	test_contract_reference_controls.SolidityReferenceControlTests.test_no_tx_origin	PASSED	0.009
7	test_contract_reference_controls.SolidityReferenceControlTests.test_timelock_is_48_hours_and_cancellable	PASSED	0.007
8	test_custody_governance.CustodyTests.test_amount_mutation_blocked	PASSED	0.121
9	test_custody_governance.CustodyTests.test_bound_intent_executes_with_three_distinct_roles	PASSED	0.055
10	test_custody_governance.CustodyTests.test_chain_mutation_blocked	PASSED	0.056
11	test_custody_governance.CustodyTests.test_custody_events_form_valid_chain	PASSED	0.059
12	test_custody_governance.CustodyTests.test_destination_mutation_blocked	PASSED	0.043
13	test_custody_governance.CustodyTests.test_duplicate_execution_blocked	PASSED	0.053
14	test_custody_governance.CustodyTests.test_executor_must_be_independent	PASSED	0.037
15	test_custody_governance.CustodyTests.test_expired_intent_blocked	PASSED	0.036
16	test_custody_governance.CustodyTests.test_fee_mutation_blocked	PASSED	0.040
17	test_custody_governance.CustodyTests.test_policy_change_requires_two_independent_approvers	PASSED	0.030
18	test_custody_governance.CustodyTests.test_policy_proposer_cannot_approve	PASSED	0.017
19	test_custody_governance.CustodyTests.test_proposer_cannot_review	PASSED	0.024
20	test_custody_governance.CustodyTests.test_unallowlisted_destination_cannot_be_reviewed	PASSED	0.018
21	test_custody_governance.GovernanceTests.test_approved_mature_proposal_executes	PASSED	0.059
22	test_custody_governance.GovernanceTests.test_canceled_proposal_never_executes	PASSED	0.056
23	test_custody_governance.GovernanceTests.test_early_execution_denied	PASSED	0.039

APPENDIX A.2

Automated Test Register - 24 to 46

#	Test ID	Status	ms
24	test_custody_governance.GovernanceTests.test_governance_events_form_valid_chain	PASSED	0.060
25	test_custody_governance.GovernanceTests.test_independent_review_required	PASSED	0.024
26	test_custody_governance.GovernanceTests.test_minimum_delay_is_48_hours	PASSED	0.023
27	test_custody_governance.GovernanceTests.test_proposal_id_binds_calldata	PASSED	0.036
28	test_custody_governance.GovernanceTests.test_reviewer_cannot_execute	PASSED	0.041
29	test_custody_governance.GovernanceTests.test_unapproved_target_rejected	PASSED	0.037
30	test_identity_provenance.IdentityTests.test_custody_plane_is_separate	PASSED	0.027
31	test_identity_provenance.IdentityTests.test_decision_is_audited	PASSED	0.033
32	test_identity_provenance.IdentityTests.test_expired_jit_denied	PASSED	0.022
33	test_identity_provenance.IdentityTests.test_high_risk_replayed_session_denied	PASSED	0.023
34	test_identity_provenance.IdentityTests.test_missing_role_denied	PASSED	0.030
35	test_identity_provenance.IdentityTests.test_non_phishing_resistant_factor_denied	PASSED	0.037
36	test_identity_provenance.IdentityTests.test_revoked_session_denied	PASSED	0.029
37	test_identity_provenance.IdentityTests.test_unknown_privilege_denied	PASSED	0.018
38	test_identity_provenance.IdentityTests.test_unmanaged_device_denied	PASSED	0.020
39	test_identity_provenance.IdentityTests.test_valid_privilege_allowed	PASSED	0.016
40	test_identity_provenance.ProvenanceTests.test_artifact_substitution_quarantined	PASSED	0.083
41	test_identity_provenance.ProvenanceTests.test_builder_substitution_quarantined	PASSED	0.053
42	test_identity_provenance.ProvenanceTests.test_commit_mismatch_quarantined	PASSED	0.045
43	test_identity_provenance.ProvenanceTests.test_environment_substitution_quarantined	PASSED	0.042
44	test_identity_provenance.ProvenanceTests.test_future_statement_quarantined	PASSED	0.051
45	test_identity_provenance.ProvenanceTests.test_provenance_decision_is_audited	PASSED	0.050
46	test_identity_provenance.ProvenanceTests.test_sbom_mismatch_quarantined	PASSED	0.041

APPENDIX A.3

Automated Test Register - 47 to 69

#	Test ID	Status	ms
47	test_identity_provenance.ProvenanceTests.test_signature_tamper_quarantined	PASSED	0.038
48	test_identity_provenance.ProvenanceTests.test_source_mismatch_quarantined	PASSED	0.036
49	test_identity_provenance.ProvenanceTests.test_valid_attestation_allowed	PASSED	0.035
50	test_network_ledger_incident.EgressTests.test_allowlisted_hostname_allowed	PASSED	0.059
51	test_network_ledger_incident.EgressTests.test_decision_is_audited	PASSED	0.038
52	test_network_ledger_incident.EgressTests.test_raw_ip_denied	PASSED	0.031
53	test_network_ledger_incident.EgressTests.test_trailing_dot_is_normalized	PASSED	0.026
54	test_network_ledger_incident.EgressTests.test_unknown_hostname_denied	PASSED	0.022
55	test_network_ledger_incident.EgressTests.test_wrong_port_denied	PASSED	0.020
56	test_network_ledger_incident.EgressTests.test_wrong_workload_denied	PASSED	0.019
57	test_network_ledger_incident.IncidentTests.test_cannot_skip_directly_to_closed	PASSED	0.023
58	test_network_ledger_incident.IncidentTests.test_failed_recovery_can_return_to_manual_control	PASSED	0.055
59	test_network_ledger_incident.IncidentTests.test_full_state_path_closes	PASSED	0.061
60	test_network_ledger_incident.IncidentTests.test_pause_does_not_affect_other_scope	PASSED	0.018
61	test_network_ledger_incident.IncidentTests.test_scoped_pause_preserves_safe_exit	PASSED	0.016
62	test_network_ledger_incident.IncidentTests.test_state_decisions_are_audited	PASSED	0.024
63	test_network_ledger_incident.LedgerTests.test_balanced_journal_allowed	PASSED	0.022
64	test_network_ledger_incident.LedgerTests.test_balances_reconstruct_from_journals	PASSED	0.025
65	test_network_ledger_incident.LedgerTests.test_duplicate_journal_rejected	PASSED	0.026
66	test_network_ledger_incident.LedgerTests.test_exact_reconciliation_matches	PASSED	0.016
67	test_network_ledger_incident.LedgerTests.test_material_difference_freezes_only_scope	PASSED	0.016
68	test_network_ledger_incident.LedgerTests.test_single_sided_journal_rejected	PASSED	0.015
69	test_network_ledger_incident.LedgerTests.test_tolerance_is_respected	PASSED	0.016

APPENDIX A.4

Automated Test Register - 70 to 89

#	Test ID	Status	ms
70	test_network_ledger_incident.LedgerTests.test_unbalanced_journal_rejected	PASSED	0.016
71	test_weni_audit_scenario.AuditLogTests.test_appended_chain_is_valid	PASSED	0.038
72	test_weni_audit_scenario.AuditLogTests.test_empty_chain_is_valid	PASSED	0.007
73	test_weni_audit_scenario.AuditLogTests.test_event_payload_is_hashed	PASSED	0.016
74	test_weni_audit_scenario.AuditLogTests.test_reordered_events_are_detected	PASSED	0.024
75	test_weni_audit_scenario.AuditLogTests.test_tampered_event_is_detected	PASSED	0.026
76	test_weni_audit_scenario.ScenarioTests.test_black_tide_audit_chain_is_valid	PASSED	0.659
77	test_weni_audit_scenario.ScenarioTests.test_black_tide_emits_cross_plane_events	PASSED	0.962
78	test_weni_audit_scenario.ScenarioTests.test_black_tide_has_no_failed_control	PASSED	0.609
79	test_weni_audit_scenario.ScenarioTests.test_black_tide_has_zero_simulated_principal_loss	PASSED	0.571
80	test_weni_audit_scenario.ScenarioTests.test_black_tide_is_contained	PASSED	0.771
81	test_weni_audit_scenario.ScenarioTests.test_black_tide_is_deterministic	PASSED	1.214
82	test_weni_audit_scenario.WeniTests.test_amount_change_returns_to_review	PASSED	0.050
83	test_weni_audit_scenario.WeniTests.test_destination_change_returns_to_review	PASSED	0.043
84	test_weni_audit_scenario.WeniTests.test_expired_request_returns_to_review	PASSED	0.034
85	test_weni_audit_scenario.WeniTests.test_hostile_untrusted_source_cannot_win	PASSED	0.019
86	test_weni_audit_scenario.WeniTests.test_no_trusted_evidence_is_rejected	PASSED	0.019
87	test_weni_audit_scenario.WeniTests.test_prepared_request_is_unsigned_and_review_required	PASSED	0.037
88	test_weni_audit_scenario.WeniTests.test_weni_cannot_sign	PASSED	0.026
89	test_weni_audit_scenario.WeniTests.test_weni_events_form_valid_chain	PASSED	0.023

APPENDIX B.1

Source Manifest - 1 to 24

#	SHA-256	Path
1	19438f822430f806abf3fddd...	src/sherlock_controls/__init__.py
2	8cca238bd01214b532038e05...	src/sherlock_controls/audit.py
3	8c088daab3e6c7169dba8b3c...	src/sherlock_controls/custody.py
4	69d2bf97fa62d3525e9260c3...	src/sherlock_controls/egress.py
5	da6a64b4ef53332ab4af7cab...	src/sherlock_controls/governance.py
6	1e02e46c361cc88bab62b174...	src/sherlock_controls/identity.py
7	f6602352e2c588990ece52da...	src/sherlock_controls/incident.py
8	7b3ef711c14beef4ae91a85c...	src/sherlock_controls/ledger.py
9	34c66c392e0bfc86edfb5d3...	src/sherlock_controls/provenance.py
10	c01c80c891102d0936df1956...	src/sherlock_controls/scenario.py
11	6cb98d4df92376c06581ef91...	src/sherlock_controls/weni.py
12	67319559958e3a6e8fac0a2...	tests/__init__.py
13	e43530bd9cc3c060a1fd153c...	tests/test_contract_reference_controls.py
14	90b11fc7913c232f8226a726...	tests/test_custody_governance.py
15	14a4c8cec31f88a02058a744...	tests/test_identity_provenance.py
16	d83fd9bacff202e96e66edd6...	tests/test_network_ledger_incident.py
17	cb8470033c04d68d9aa3b68b...	tests/test_weni_audit_scenario.py
18	76c7febdb65080faf3fb176d...	policies/custody_policy.json
19	08ff56d353f0811c67d40e0a...	policies/egress_policy.json
20	f28eb217ce31bf06c80642e2...	policies/identity_policy.json
21	cc3152c3587dbc7f9e9acd12...	policies/incident_policy.json
22	13b95ec35312ca52c31c46f6...	schemas/audit-event.schema.json
23	60874df7c933a32e5cf0df2b...	schemas/revalidation-result.schema.json
24	ff9ce8329c0667f5be3a9521...	contracts/BoundWithdrawalIntent.sol

APPENDIX B.2

Source Manifest - 25 to 33

#	SHA-256	Path
25	231fad1675764299a089f4d7...	contracts/EvidenceHashRegistry.sol
26	b181cf0a02897974270b76fb...	contracts/SherlockTimelockGuard.sol
27	b1b521d42bd0ec8c030f57e0...	diagrams/architecture.mmd
28	f9d8b8422a7ad5f70d621e3b...	diagrams/attack-sequence.mmd
29	04ddf8730e3382d43717beb4...	diagrams/incident-state.mmd
30	d7270481bede7ef2a030c84d...	diagrams/weni-boundary.mmd
31	97fe3542ced054673487c879...	scripts/run_revalidation.py
32	39fe8d86e29c36147a687c67...	README.md
33	8607c44a235e8e71468d2cff...	pyproject.toml

FINAL

Final Remediation Verification Conclusion

All ten findings from SHL-WCF-AE-2026-0814 are resolved and retested inside the SHERLOCK isolated revalidation environment.

Decision	Value
Findings resolved	10 / 10
Open findings	0
Automated tests	89 / 89 PASSED
Scenario outcome	CONTAINED
Simulated principal loss	\$0
Audit chain	VALID
Source root	d3a843604a213fb8aed9062c4a0193c819401985390a64ac72f4a3a78e2d65fa
Assurance	O1-ISOLATED
Revalidation status	VERIFIED

SHERLOCK closing opinion

The remediated design demonstrates fail-closed behavior across identity, release integrity, network egress, custody, governance, ledger, WENI and incident recovery. Every adversary mutation in the defined scenario is independently interrupted and evidenced. No open action remains inside this scope.

Production adoption requires mapping these exact controls and hashes to the live environment. That mapping is a deployment assurance activity, not an unresolved defect in this isolated remediation verification.

SHERLOCK

Internal Security Assurance

SHL-WCF-RV-2026-0814 | v2.0 | 14 August 2026